

Power BI Questions and Answers

1. How do you calculate the number of days between two dates using DAX? Dates be 'Start_Date' and 'End_Date' and table name as 'tbl'.

Answer: You can calculate the number of days between two dates using the DAX formula:

```
DaysBetween = DATEDIFF(tbl[Start_Date], tbl[End_Date], DAY)
```

This DAX function computes the difference in days between the 'Start_Date' and 'End_Date' columns in the 'tbl' table.

2. What are the different levels of filters in Power BI?

Answer: Power BI provides several levels of filters:

1. Report Level Filters: These filters apply to all the pages within the report.
2. Page Level Filters: These filters apply to all the visuals on a specific report page.
3. Visual Level Filters: These filters apply only to the selected visual on a report page.

Each level of filtering allows for more granular control of the data being visualized.

3. Difference between Power Query and Power Pivot

Answer: Power Query is used for data extraction, transformation, and loading (ETL), allowing users to import data from various sources, clean, and reshape it. Power Pivot, on the other hand, is used for creating data models, establishing relationships, and performing calculations using DAX for

analysis.

Power Query is typically used before data is loaded into Power Pivot, providing clean and transformed data for further modeling and analysis.

4. How can you calculate the sum of sales for a specific category using DAX? Table name 'orders', columns are: 'Sales', 'Category'

Answer: To calculate the sum of sales for a specific category using DAX, you can use the following formula:

```
SalesForCategory = CALCULATE(SUM(orders[Sales]), orders[Category] = 'SpecificCategory')
```

Replace 'SpecificCategory' with the actual category you want to filter by.

5. Given a table named Sales with columns Product, Date, and Revenue, write a DAX measure called 'AverageDailyRevenue' that calculates the average daily revenue for a specific product over the last 30 days, considering only weekdays (Monday to Friday).

Answer: You can create the 'AverageDailyRevenue' measure with the following DAX formula:

```
AverageDailyRevenue =  
AVERAGEX(  
    FILTER(  
        ADDCOLUMNS(  
            CALENDAR(MIN(Sales[Date]), MAX(Sales[Date])),  
            "WeekDay", WEEKDAY([Date], 2)
```

```
),  
[WeekDay] <= 5 && Sales[Date] >= TODAY() - 30  
),  
CALCULATE(SUM(Sales[Revenue]), Sales[Product] = "SpecificProduct")  
)
```

Replace 'SpecificProduct' with the product you're interested in.

32. Differentiate between Drill Through and Drill Down.

Answer: Drill Down allows you to explore data in more detail within the same visual by moving from higher-level summary data to lower-level detailed data.

Example: In a bar chart showing sales by year, you can drill down to see sales by quarter, and further down to see sales by month.

Drill Through allows users to navigate from one report page to another to see detailed information related to a specific data point.

Example: Clicking on a specific product in a sales summary report might drill through to another page that shows detailed transactions for that product.

33. Differentiate between Implicit and Explicit Measures.

Answer: Implicit measures are created automatically by Power BI when you drag and drop a numeric field into a visual. They are based on default aggregations (like SUM, AVERAGE) and are not reusable across different visuals.

Example: Dragging the 'SalesAmount' column into a report automatically creates an implicit measure that sums the sales amounts.

Explicit measures are created by the user using DAX. They are defined in the data model, and you have full control over the calculation logic. Explicit measures are reusable across different visuals and reports.

Example: A DAX measure like 'TotalSales = SUM(Sales[SalesAmount])' is an explicit measure.

40. Calculate the average annual growth rate in health expenditure for the year 2020. Create a suitable chart to compare health expenditure across different countries/regions for the year 2020.

Answer: To calculate the Average Annual Growth Rate (AAGR) for 2020:

AAGR_2020 =

VAR PrevYearSpending = CALCULATE(SUM(Sales[HealthExpenditure]), Sales[Year] = 2019)

VAR CurrentYearSpending = CALCULATE(SUM(Sales[HealthExpenditure]), Sales[Year] = 2020)

RETURN

DIVIDE(CurrentYearSpending - PrevYearSpending, PrevYearSpending)

Use a clustered bar or column chart to visualize the 'HealthExpenditure' for 2020 across different countries/regions. Add data labels to indicate the values clearly, and use a slicer to filter by country or region if needed.